Project_02

2023-03-02

```
#install.packages("tidyverse")
#install.packages("corrplot")
#install.packages("caret")
```

Load data

The data file provided is an .xlsx file which only returns a single column database. So we had to modify a .csv file while changing the separator used in sep=“,”. This gave us the file “bank-full.csv”.

```
library(tidyverse)
```

```
## -- Attaching packages --------------------------------------- tidyverse 1.3.2 --
## v ggplot2 3.4.1     v purrr   1.0.1
## v tibble  3.1.8     v dplyr   1.1.0
## v tidyr   1.3.0     v stringr 1.5.0
## v readr   2.1.4     v forcats 1.0.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
```

```
data <- read_csv("bank-full.csv", show_col_types = FALSE)

head(data)
```

```
## # A tibble: 6 x 17
##     age job    marital educa~1 default balance housing loan  contact   day month
##   <dbl> <chr>  <chr>   <chr>   <chr>     <dbl> <chr>   <chr> <chr>   <dbl> <chr>
## 1    58 manag~ married tertia~ no         2143 yes     no    unknown     5 may
## 2    44 techn~ single  second~ no           29 yes     no    unknown     5 may
## 3    33 entre~ married second~ no            2 yes     yes   unknown     5 may
## 4    47 blue-~ married unknown no         1506 yes     no    unknown     5 may
## 5    33 unkno~ single  unknown no            1 no      no    unknown     5 may
## 6    35 manag~ married tertia~ no          231 yes     no    unknown     5 may
## # ... with 6 more variables: duration <dbl>, campaign <dbl>, pdays <dbl>,
## #   previous <dbl>, poutcome <chr>, y <chr>, and abbreviated variable name
## #   1: education
```

Reading the data gives us a table 45211 and 17 variables or features.

Exploratory Data and Analysis

```
summary(data)
```

Data describe

```
##       age            job              marital           education
##  Min.   :18.00   Length:45211       Length:45211       Length:45211
```

```
##  1st Qu.:33.00   Class :character   Class :character   Class :character
##  Median :39.00   Mode  :character   Mode  :character   Mode  :character
##  Mean   :40.94
##  3rd Qu.:48.00
##  Max.   :95.00
##    default            balance         housing             loan
##  Length:45211       Min.   : -8019   Length:45211       Length:45211
##  Class :character   1st Qu.:    72   Class :character   Class :character
##  Mode  :character   Median :   448   Mode  :character   Mode  :character
##                     Mean   :  1362
##                     3rd Qu.:  1428
##                     Max.   :102127
##    contact               day           month              duration
##  Length:45211       Min.   : 1.00   Length:45211       Min.   :   0.0
##  Class :character   1st Qu.: 8.00   Class :character   1st Qu.: 103.0
##  Mode  :character   Median :16.00   Mode  :character   Median : 180.0
##                     Mean   :15.81                      Mean   : 258.2
##                     3rd Qu.:21.00                      3rd Qu.: 319.0
##                     Max.   :31.00                      Max.   :4918.0
##     campaign          pdays          previous          poutcome
##  Min.   : 1.000   Min.   : -1.0   Min.   :  0.0000   Length:45211
##  1st Qu.: 1.000   1st Qu.: -1.0   1st Qu.:  0.0000   Class :character
##  Median : 2.000   Median : -1.0   Median :  0.0000   Mode  :character
##  Mean   : 2.764   Mean   : 40.2   Mean   :  0.5803
##  3rd Qu.: 3.000   3rd Qu.: -1.0   3rd Qu.:  0.0000
##  Max.   :63.000   Max.   :871.0   Max.   :275.0000
##       y
##  Length:45211
##  Class :character
##  Mode  :character
##
##
##
```

```
print(table(sapply(data, class)))
```

```
##
## character   numeric
##        10         7
```

```
print(table(sapply(data, typeof)))
```

```
##
## character    double
##        10         7
```

The database contains 10 categorical variables and 7 numerical variables.

```
sum(is.na(data))
```

Missing values checking

```
## [1] 0
```

No missing values in the database.

```r
data$y <- as.numeric(factor(data$y))

data$y <- data$y-1
```

Target encoding

Data splitting

```r
set.seed(101) # Set Seed so that same sample can be reproduced in future also

data <- select(data, -c("day","month"))

sample <- sample.int(n = nrow(data), size = floor(.85*nrow(data)), replace = F) # Now Selecting 85% of

train_data <- data[sample, ]
test_data  <- data[-sample, ]
```

Modeling

```r
formula <- y~.
model <- glm(formula, data = train_data, family = 'binomial')
summary(model)
```

```
##
## Call:
## glm(formula = formula, family = "binomial", data = train_data)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -5.6908  -0.4074  -0.2716  -0.1612   3.0758
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)        -2.703e+00  1.824e-01 -14.824  < 2e-16 ***
## age                 1.445e-03  2.333e-03   0.619 0.535659
## jobblue-collar     -4.038e-01  7.753e-02  -5.209 1.90e-07 ***
## jobentrepreneur    -3.983e-01  1.324e-01  -3.007 0.002635 **
## jobhousemaid       -4.522e-01  1.419e-01  -3.186 0.001442 **
## jobmanagement      -2.090e-01  7.749e-02  -2.697 0.006999 **
## jobretired          3.866e-01  1.012e-01   3.819 0.000134 ***
## jobself-employed   -4.073e-01  1.195e-01  -3.409 0.000652 ***
## jobservices        -2.791e-01  8.924e-02  -3.128 0.001761 **
## jobstudent          5.092e-01  1.155e-01   4.407 1.05e-05 ***
## jobtechnician      -2.506e-01  7.274e-02  -3.445 0.000571 ***
## jobunemployed      -2.142e-01  1.188e-01  -1.803 0.071393 .
## jobunknown         -3.404e-01  2.510e-01  -1.356 0.174998
## maritalmarried     -1.208e-01  6.298e-02  -1.918 0.055167 .
## maritalsingle       2.171e-01  7.145e-02   3.039 0.002375 **
## educationsecondary  1.293e-01  6.841e-02   1.890 0.058726 .
## educationtertiary   3.853e-01  7.932e-02   4.857 1.19e-06 ***
## educationunknown    2.237e-01  1.093e-01   2.046 0.040777 *
## defaultyes         -1.207e-01  1.763e-01  -0.685 0.493320
```

```
## balance             1.910e-05  5.420e-06   3.524 0.000426 ***
## housingyes         -7.777e-01  4.287e-02 -18.141  < 2e-16 ***
## loanyes            -5.750e-01  6.325e-02  -9.091  < 2e-16 ***
## contacttelephone   -1.236e-01  7.871e-02  -1.570 0.116343
## contactunknown     -1.166e+00  6.305e-02 -18.487  < 2e-16 ***
## duration            4.025e-03  6.809e-05  59.121  < 2e-16 ***
## campaign           -1.106e-01  1.072e-02 -10.312  < 2e-16 ***
## pdays               3.790e-04  3.225e-04   1.175 0.239903
## previous            2.986e-02  9.837e-03   3.035 0.002404 **
## poutcomeother       2.535e-01  9.427e-02   2.689 0.007160 **
## poutcomesuccess     2.339e+00  8.626e-02  27.117  < 2e-16 ***
## poutcomeunknown    -1.133e-01  1.008e-01  -1.124 0.261094
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 27717  on 38428  degrees of freedom
## Residual deviance: 19228  on 38398  degrees of freedom
## AIC: 19290
##
## Number of Fisher Scoring iterations: 6
```

It is necessary to separate the database into a training and test sample because otherwise it would be impossible to evaluate the real performance of this model.

Training on test data ! Yes or No ? No need to train the model on the test database. To avoid overfitting the data on the test database and thus compromising our analysis of the real performance of the model. There is always the specific case where by studying the learning curve of our model we realize that it is still learning. In this case it would be interesting to complete the training database to gain in performance.

Make prediction on test data

```
predicted <- predict(model, select(test_data, -c("y")), type="response")
```

Evaluate model

```
library(caret)
```

```
## Le chargement a nécessité le package : lattice
```

```
##
## Attachement du package : 'caret'
```

```
## L'objet suivant est masqué depuis 'package:purrr':
##
##     lift
```

```
confusionMatrix(as.factor(test_data$y), as.factor(round(predicted)))
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##          0 5849  135
```

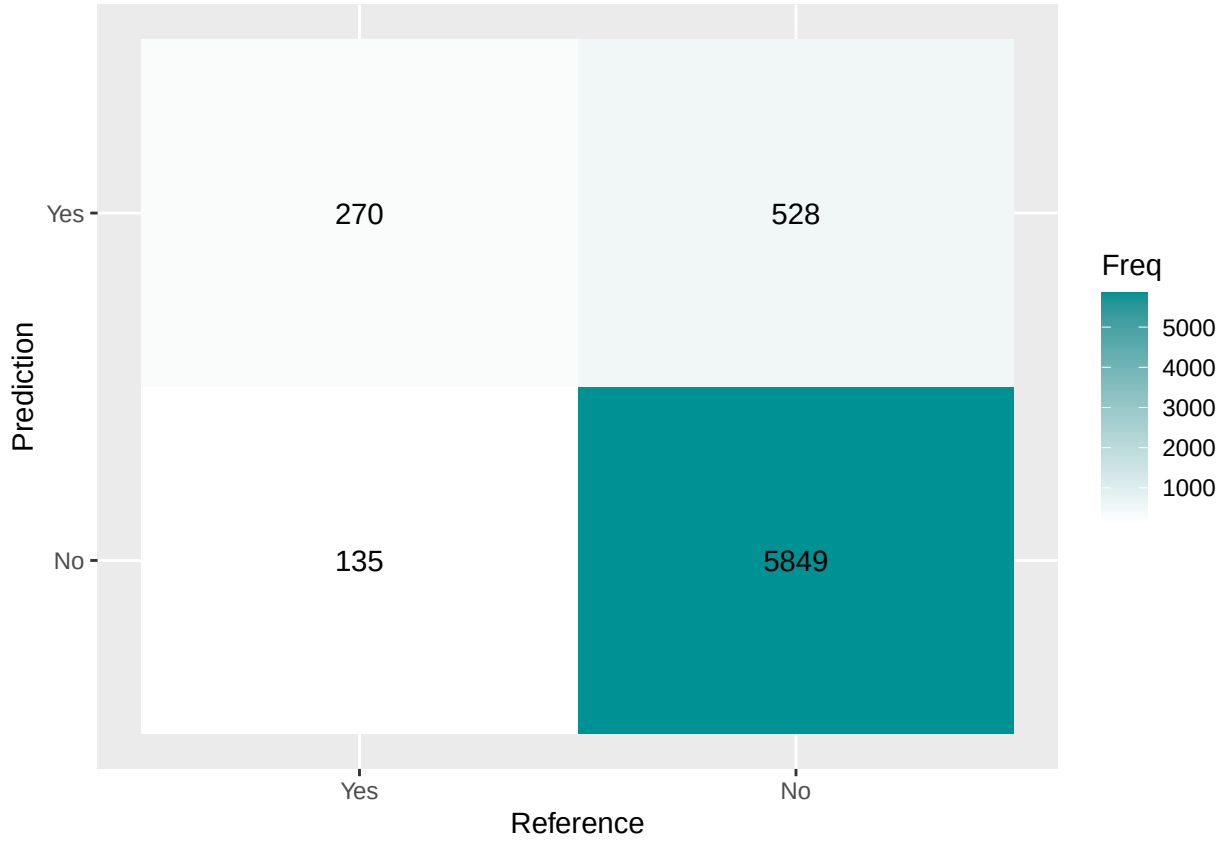
```
##          1  528  270
##
##                Accuracy : 0.9022
##                  95% CI : (0.8949, 0.9092)
##     No Information Rate : 0.9403
##     P-Value [Acc > NIR] : 1
##
##                   Kappa : 0.4015
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.9172
##             Specificity : 0.6667
##          Pos Pred Value : 0.9774
##          Neg Pred Value : 0.3383
##              Prevalence : 0.9403
##          Detection Rate : 0.8624
##    Detection Prevalence : 0.8823
##       Balanced Accuracy : 0.7919
##
##        'Positive' Class : 0
##
```

Confusion matrix

```
cm <- confusionMatrix(as.factor(test_data$y), as.factor(round(predicted)), dnn = c("Reference", "Predic

plt <- as.data.frame(cm$table)
plt$Prediction <- factor(plt$Prediction, levels=rev(levels(plt$Prediction)))

ggplot(plt, aes(Prediction,Reference, fill= Freq)) +
        geom_tile() + geom_text(aes(label=Freq)) +
        scale_fill_gradient(low="white", high="#009194") +
        labs(x = "Reference",y = "Prediction") +
        scale_y_discrete(labels=c("No","Yes")) +
        scale_x_discrete(labels=c("Yes","No"))
```


Yes
No
Prediction
270
528
135
5849
Yes
No
Reference
Freq
5000
4000
3000
2000
1000